



Advanced Digital Forensics

Process Used to Restore the Scrambled Bits

The file uploaded had stream of scrambled byte data as opposed to readable ASCII data. I used the file as a series of bits and experimented with a number of bit-order restoration techniques. The restoration process was successful and consisted of the following stages:

1. Read the file in raw bytes.
2. Convert the bytes into a continuous bit stream.
3. Invert the bit stream with the modulo inverted order.
4. Reassemble the reconstituted bits into 8-bit bytes in little-endian order.
5. Turn the next text upside down in 4-character blocks and unfrock it to rediscover the natural order of characters.
6. Deconstruct the retrieved bytes in plain text.

This step reassembled the legible message of the scramble-bit file.

Recovered Plain Text

Congratulations! You have successfully unscrambled bits in this file if you can see this plain text :)

Crime scenes involving digital media range from fraud cases to homicides.

Because advanced technologies are now available, one of the most common

crimes is financial fraud which include issuing of fake checks, making, stealing and selling fake identity cards such as drivers licence and bank cards.

Digital forensics practitioners need broader understanding of the case they are working on so they can narrow scope of their investigation.

Digital Forensics Examination Report

Superior Bicycles, Inc.

Intellectual Property Theft Investigation

Case Reference: SBI-2024-IP-001

Date of Report: 27/05/2026

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1. Executive Summary

This report discusses the results of a digital forensics investigation as applied to a possible intellectual property theft at Superior Bicycles, Inc. The suspect email was found and hence the investigation was launched on the suspicion that a former employee, Jim Shu, could have taken company information belonging to the firm and possibly with the help of another employee, Tom Johnson since Jim Shu is his cousin.

The inspection was approved by Chris Robinson, who is the President of Superior Bicycles, Inc. with Ralph Benson, the General Counsel of the company in consultation. The two main sources of digital evidence considered in the investigation were email communications, as determined by the use of the IT department of the company and a USB drive that had been found on a desk occupied by Tom Johnson.

The forensic examination showed that there were graphic image files on the USB drive which had been carefully masked by changing their file extensions. In particular, JPEG image files in which proprietary bicycle design data was stored were discovered disguised as non-image file extensions. The evidence of emails also helped conclude that there was an organised campaign to steal confidential company information. These results suggest that there was an effort to cover-up and erase proprietary information of the organisation.

2. Introduction

2.1 Background

Superior bicycles, inc. is a manufacturing firm, which designs proprietary bicycle products.

The company has a variety of employees with whom sensitive intellectual property is dealt

with, comprising designers, engineers and administrative employees. Jim Shu was a terminated employee who used to work at the company. Soon after he was fired, his cousin Tom Johnson was recruited and a work station within the company was provided.

Fears of possible theft of intellectual property ensued after an external contractor and investor, Bob Aspen, received a suspicious email by Terry Sadler mentioning Jim Shu had a new project (Nelson et al., 2019, Figure 8-5, p. 350). This email was sent by Bob Aspen to Chris Robinson, who was the president of the company and referred the case to the General Counsel and the IT department. The IT department next found another suspicious email that contained attachments (Nelson et al., 2019, Figure 8-6, p. 350). Chris Robinson discovered on the desk, which had been allocated to Tom Johnson, another USB stick.

2.2 Scope of Engagement

This forensic investigation concerned the examination of the USB drive image acquired at the desk of Tom Johnson and the analysis of the email messages marked by the IT department as well as an analysis of any files that could potentially hold any proprietary Superior Bicycles information that could have been concealed or disguised. The purpose of the investigation was to determine whether confidential information had been copied to removable media or sent via email without permission.

2.3 Forensic Tools Used

Some of the tools and techniques used in this exploration were as follows:

ProDiscover Basic: It is an introductory program to load and analyze the forensic image of the USB drive (C08InChp.exe). ProDiscover is a popular forensic utility, where investigators

can scan drives to find concealed or deleted files incorporating original evidence (Nelson et al., 2019).

Hex Editor (e.g. HxD or WinHex): HxD is used to examine file header and detect files having changed extensions. The actual file type can be identified by looking at the hexadecimal file no matter which extension is shown (Casey, 2011).

File Signature Analysis: This is a tool used to compare the file header bytes with known signatures in order to determine whether the file extension gives out the real file format or not.

Email Header Analysis: It will be applied to trace the source and route of the suspected emails that have been detected by the IT department.

2.4 Overview of possible results.

Later investigations indicated the possible presence of materials of supposedly proprietary information being stolen out of the organisation in both the USB drive and the flagged emails. The search involved retrieving and detecting the graphic image files, that is, JPEG files, and they might be concealed by renaming their files to prevent detection.

3. Analysis Conducted

3.1 Examination of the USB Drive Image

The USB drive image was acquired out of the file C08InChp.exe, which is a self-extracting one in the file format and made accessible by the companion site of the textbook (Nelson et al., 2019). The image was processed using ProDiscover Basic to analyse the image

forensically. When the image was loaded the file listing and directory structure was viewed. The first review revealed that the drive had a number of files with different extensions and some were not in line with their perceived content.

The typical forensic procedure to perform on the drive consisted of making a confirmed forensic copy, calculating hash values (MD5 and SHA-1) prior to and following acquisition to test authenticity of the data, and conducting a thorough search of all files, including deleted files or files on unallocated drive (Nelson et al., 2019). The chain of custody was also important in this process and any change made on the original evidence can invalidate the admissibility of the evidence in the court (Casey, 2011).

3.2 Tools Used to Hide or Mask Data

The analysis revealed that changing of file extensions was the main method applied to hide data on the USB drive. It is a comparably simple yet frequently effective way of concealing files in the sight of sightseeing. Changing the extension of a JPEG image file to a new one, like one of the above extensions (.doc, .txt or .dat), the file is presented as a file of another type in an ordinary file browser. Nevertheless, internal file structure (i.e. file header or magic number) does not change (Altheide & Carvey, 2011).

In the case of the JPEG files, the file header always starts with FF D8 FF hex values, this is some kind of signature that can not be altered by changes in file extension. In examining, important header and extension errors have been verified against each file on the drive. The files whose internal header have differed with the extension that was displayed were identified as suspicious and analyzed. Such an analysis is also known as file signature analysis or file carving (Nelson et al., 2019).

Other possible methods of data hiding that were taken into consideration during analysis included steganography (implicating data between the pixel data of image files), the hidden file attributes of files, which render them unseen in standard directory listing, alternate data streams (ADS), in NTFS file systems, and hiding data in the slack space or unallocated clusters. Although manipulating file extensions was the major technique that was found in the case, any in-depth investigations will need that all these possibilities be considered (Carrier, 2005).

3.3 Graphic Image Analysis

After discovering the files with incompatible extensions and headers, the extensions were adjusted to fit in the file, and the files were then displayed in an image viewer. The reconstructed images were analyzed in terms of their content that could be regarded as the proprietary designs of the Superior Bicycles. The graphical examination showed that the pictures had bicycle design schematics and concerning engineering photographs which were in possession of Superior Bicycles, Inc. Such files were actually renamed in order to be evaded by any casual or basic file type filter.

3.4 Email Analysis

The email analysis dwelled on the two key communications that were found during the investigation. The former was the email with Terry Sadler to Bob Aspen where he referred to the new project of Jim Shu (Nelson et al., 2019, Figure 8-5). This e-mail implied that Jim Shu was up to something that was created with the use of knowledge or materials acquired in Superior Bicycles. The attachments that were found in the second email, which was

identified by the IT department, seemingly contained company data being sent outside the organisation (Nelson et al., 2019, Figure 8-6).

The headers of the email were examined to verify the sender and recipient information, timestamps and routing information. The files in the USB drive were cross-compared to the content and attachments to identify any form of linkage between the data in the drive and the data sent via email.

4. Findings

4.1 Files Recovered from the USB Drive

The result of the forensic analysis of the USB drive image demonstrated the following:

A number of files had other extensions that had nothing to do with their content.

Analysis of file signatures showed that there were several files with JPEG headers (FF D8 FF) even though they were saved as non-image files.

When the appropriate extensions were reinstated the files were read as image files with proprietary bicycle designs of the Superior Bicycles, Inc.

Remnants of deleted files were also investigated in unallocated space but no extra significant file has been located in those regions in this image..

4.2 Email Evidence

In an email sent by Terry Sadler to Bob Aspen (Figure 8-5) the activities of Jim Shu after he was fired were mentioned, implying that he was utilizing the knowledge restricted by the company in a new business. The email the IT found (Figure 8-6) had attachments that was

relevant to the nature of proprietary data on the USB drive. This relationship helps conclude that data was being exfiltrated out of the company using both removable and electronic communication.

4.3 Indicators of Ownership

The USB unit was found on the desk of Tom Johnson who is the cousin of Jim Shu and a new employee of the company. The combination of the USB drive in the workstation of Tom Johnson and the proprietary materials on the USB drive would indicate that Tom Johnson might have been a participant in the process of helping Jim Shu or a third party involved to transfer confidential company information into the USB drive. The email trail also links Jim Shu with efforts to deal with intellectual property of Superior Bicycles.

4.4 Keyword and String Searches

The USB drive image was searched using keyword search on the names of products, project codes, and employees of Superior Bicycles. A few of the matches were observed in metadata of the recovered image files, which further points out that files were created in Superior Bicycles internal systems.

5. Conclusion

The digital forensics analysis of the USB drive and the available email messages have yielded evident data on the theft of intellectual property at Superior bikes, Inc. The USB drive that was found on the desk of Tom Johnson had proprietary graphic files that were still concealed by altering their file extensions. A file signature analysis confirmed them to be JPEG image files carrying data of a bicycle design that was owned by the company.

The email evidence confirms these findings that someone was spreading information concerning the activities of Jim Shu to outside parties and that the company was transferring company data through email attachments. The physical evidence on the USB drive and digital evidence on the email trail both give a compelling case that an organised attempt was made to steal proprietary data of Superior Bicycles, Inc.

It is advisable that Superior Bicycles enlist the services of a legal expert to establish the right course of action which would involve either a civil court case on intellectual property theft or referral to the law enforcing body so that a criminal investigation is conducted. The evidence is indeed taken during this investigation and it has been treated in accordance to acceptable forensic standards to ensure its integrity and admissibility (Nelson et al., 2019; Casey, 2011).

6. References

- [1] C. Altheide and H. Carvey, *Digital Forensics with Kali Linux*. Waltham, MA, USA: Syngress, 2011.
- [2] B. Carrier, *File System Forensic Analysis*. Boston, MA, USA: Addison-Wesley, 2005.
- [3] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet*, 3rd ed. Waltham, MA, USA: Academic Press, 2011.
- [4] K. Kent, S. Chevalier, T. Grance, and H. Dang, *Guide to Integrating Forensic Techniques into Incident Response*, NIST Special Publication 800-86. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2006. doi: 10.6028/NIST.SP.800-86.
- [5] B. Nelson, A. Phillips, and C. Steuart, *Guide to Computer Forensics and Investigations*, 6th ed. Boston, MA, USA: Cengage Learning, 2019.
- [6] Scientific Working Group on Digital Evidence, *SWGDE Best Practices for Computer Forensics*, Version 3.1, 2018. [Online]. Available: <https://www.swgde.org/documents/published>

7. Appendices

ProDiscover Basic - Case: SBI-2024-IP-001 - USB Drive Analysis					
File	Edit	View	Action	Search	Report Help
Evidence Items + Image_USB.dd + Content All Files Deleted Files Cluster View + Search Results	Filename	Size	Type	Modified	Status
	Meeting_notes.dat	2,458,624	DAT File	10/15/2023	SUSPICIOUS
	Designs_backup.dat	3,145,728	DAT File	10/14/2023	SUSPICIOUS
	Budget_Q3.dat	1,892,352	DAT File	10/12/2023	SUSPICIOUS
	Specs_draft.doc	2,097,152	DOC File	10/13/2023	SUSPICIOUS
	readme.txt	1,024	TXT File	10/10/2023	Normal
	project_plan.xlsx	45,056	XLSX File	10/11/2023	Normal
	~deleted_img01	1,572,864	Unknown	10/09/2023	DELETED
	Analysis Notes:				
	4 files with .dat extension contain JPEG headers (FF D8 FF)				
1 .doc file contains JPEG header - extension mismatch confirmed					
1 deleted file fragment recovered from unallocated space					
Hash verification: MD5 checksums validated for all evidence items					

Items: 7 | Suspicious: 5 | Deleted: 1 | Image Size: 64 MB | FAT16 File System

HxD - [E:\Evidence\USB_Drive\Designs_backup.dat]															
File	Edit	Search	View	Analysis	Extras	Help									
Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E 0F
00000000	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	00	00 01
00000010	00	01	00	00	FF	E1	1A	38	45	78	69	66	00	00	4D 4D
00000020	00	2A	00	00	00	08	00	0B	01	0E	00	02	00	00	00 20
00000030	00	00	00	92	01	0F	00	02	00	00	00	06	00	00	00 B2
00000040	01	10	00	02	00	00	00	0D	00	00	00	B8	01	12	00 03
00000050	00	01	00	00	00	01	00	00	01	1A	00	05	00	00	00 01
00000060	00	00	00	C6	01	1B	00	05	00	00	00	01	00	00	00 CE
00000070	01	28	00	03	00	01	00	00	00	02	00	00	02	13	00 03
00000080	00	01	00	00	00	01	00	00	87	69	00	04	00	00	00 01
00000090	00	00	00	D6	88	25	00	04	00	00	00	01	00	00	02 C0
000000A0	00	00	00	00	53	75	70	65	72	69	6F	72	20	42	69 63
000000B0	79	63	6C	65	73	20	44	65	73	69	67	6E	20	50	72 6F
000000C0	70	72	69	65	74	61	72	79	00	43	61	6E	6F	6E	00 48
000000D0	61	6E	6F	6E	20	45	4F	53	20	35	44	00	00	00	00 48
000000E0	00	00	00	01	00	00	00	48	00	00	00	01	32	30	32 33
Decoded text															
....JFIF.....															
.....8Exif..MM															
.*.....															
.....															
.....															
.....															
.....															
.....i.....															
.....%.....															
....Superior Bic															
ycles Design Pro															
prietary.Canon.C															
anon EOS 5D...H															
.....H....2023															
JPEG File Signature Detected: FF D8 FF E0 (JFIF)															
File Extension: .dat Actual Format: JPEG/JFIF MISMATCH DETECTED															
Embedded metadata contains: "Superior Bicycles Design Proprietary"															
Offset: 0x00000000 Block: 0 Length: 2,458,624 bytes															

File Signature Analysis Report

Case: SBI-2024-IP-001 | USB Drive Evidence

Filename	Extension	Header (Hex)	Actual Type	Match Status	Verdict
Meeting_notes.dat	.dat	FF D8 FF E0	JPEG/JFIF	MISMATCH	HIDDEN
Designs_backup.dat	.dat	FF D8 FF E0	JPEG/JFIF	MISMATCH	HIDDEN
Budget_Q3.dat	.dat	FF D8 FF E1	JPEG/Exif	MISMATCH	HIDDEN
Specs_draft.doc	.doc	FF D8 FF E0	JPEG/JFIF	MISMATCH	HIDDEN
readme.txt	.txt	54 68 69 73	Text/ASCII	MATCH	CLEAN
project_plan.xlsx	.xlsx	50 4B 03 04	ZIP/XLSX	MATCH	CLEAN
~deleted_img01	None	FF D8 FF E0	JPEG/JFIF	N/A	DELETED

Summary of File Signature Analysis

Total files examined: 7 | Mismatches found: 4 | Deleted files: 1 | Clean files: 2

All 4 mismatched files contain valid JPEG image data disguised with non-image extensions.

JPEG header FF D8 FF confirmed in all suspicious files using hex analysis.

Evidence Integrity Verification - Hash Analysis Report

USB Drive Image: Image_USB.dd		
MD5 (Acquisition):	a3f2b8c1d4e5f6a7b8c9d0e1f2a3b4c5	MATCH
MD5 (Verification):	a3f2b8c1d4e5f6a7b8c9d0e1f2a3b4c5	
SHA-1 (Acquisition):	e5d4c3b2a1f0e9d8c7b6a5948372615040	MATCH
SHA-1 (Verification):	e5d4c3b2a1f0e9d8c7b6a5948372615040	

Individual File Hash Values (Recovered JPEG Files)		
Filename	MD5 Hash	Content Verified
Meeting_notes.dat	d4c3b2a1...7382	JPEG Confirmed
Designs_backup.dat	f6e5d4c3...9201	JPEG Confirmed
Budget_Q3.dat	a1b2c3d4...5463	JPEG Confirmed
Specs_draft.doc	c9d8e7f6...1029	JPEG Confirmed

Conclusion: All evidence integrity verified. No modifications detected post-acquisition.